

Assignment 1 Memo: Corpus Construction and Characterization

Buildings, Construction & Accessibility | Pittsburgh 311 | NL(X) and LLM, Group 3 | Mahika Gunjkar

My corpus is 260 Pittsburgh 311 requests pulled from the WPRDC archive and narrowed to four categories: Building Maintenance, Construction Issues, Permits, and Accessibility, with 65 records each. Every record is mixed, meaning it has a plain-language description in `raw_text` and the structured fields in `table_json`. The extraction task asks the model to pull five fields from each record: `issue_category`, `request_type`, `responsible_department`, `neighborhood`, and `entities`.

1. Human vs. SLM (Phi)

I labelled 25 records by hand and ran the same 25 through Phi-4-mini with the same schema, then compared field by field. The table shows how often Phi matched my labels.

Field	Agreement	Notes
<code>request_type</code>	1.00	copied straight from the text, no errors
<code>responsible_department</code>	1.00	copied straight from the text, no errors
<code>neighborhood</code>	1.00	copied straight from the text, no errors
<code>issue_category</code>	0.40	10 of 25; the errors are described below
<code>entities</code>	F1 0.41	P 0.42 / R 0.41; exact-set match near 0

Phi did well on the three fields that only ask it to copy text out of the record. It got `request_type`, `responsible_department`, and `neighborhood` right every time, and it never produced malformed JSON. Each record took about a minute to run on CPU.

Where it struggled was `issue_category`, which it only got right 10 out of 25 times, and the mistakes were not random. Phi called all ten of my permits records construction, and all four of my accessibility records building_maintenance. It never predicted permits or accessibility at all. Basically it flattened my four categories into two: anything about work without a permit became construction, and anything about sidewalks, curb ramps, or city steps became building maintenance. The only building_maintenance record it missed was a Vacant Building, which it also filed under construction.

Entities is harder to score. If I make Phi's list match mine exactly, agreement is basically zero, but the token-level F1 is 0.41, so there is real overlap. Some of that gap is on my labelling rule as much as on Phi, since I counted the neighborhood, street, and department as entities and Phi chose a somewhat different set. That is also why the all-fields-correct number is 0%, since a single entity mismatch fails the whole record, so the per-field numbers give a fairer picture.

2. Recovery Strategy

The recurring problem I went after was the `issue_category` confusion. My revised prompt leaned on the output format, telling the model it had to return exactly one of the four category values and nothing else. It did not help. Accuracy actually dropped a little, from 0.40 to 0.36, and no records got fixed.

Looking back, that makes sense. Phi's answers were already valid category values, since the schema-violation rate was zero the whole time, so the problem was never the format. The model just did not know that a request about work without a permit belongs under permits rather than construction. A prompt that polices the format cannot fix a meaning problem. What I would try next is spelling out what each category means, for example telling it that unpermitted work goes under permits and that sidewalks and ramps go under accessibility.

3. Cluster Reading

K-means gave me 6 clusters. The silhouette score is low, 0.067, which I think is just because the records are short and follow the same template, so they all sit close together in embedding space. The clusters line up almost perfectly with request types, which tells me the corpus does not have much structure beyond its own category labels. Here is how I read each one:

Cluster	n	Reading
0	33	Sidewalk, curb-ramp, and city-steps requests (accessibility)
1	61	Construction-site maintenance
2	25	Unpermitted work (electrical and HVAC)
3	15	Blocked or closed sidewalks (DOMI)
4	85	Building maintenance and building-without-permit (largest)
5	41	Broken sidewalks and fire-safety systems

The interesting part is how this lines up with Phi's mistakes. The categories Phi got wrong are exactly the ones that fill these clusters. The permit-heavy clusters (4 and 2) and the sidewalk clusters (0, 3, and 5) are where all the misclassifications live, because those are the permits and accessibility records Phi will not label correctly. The one cluster it handles fine is cluster 1, construction-site maintenance, and that is only because those records really are construction, so its usual construction guess happens to be right.

4. Corpus Limitations

There are a few things this corpus cannot do. It comes from a single source and a single document type, so it cannot say anything about permit outcomes, inspections, or costs. The text is generated from fields rather than written by residents, so it reads the same way every time and does not capture how people actually report problems. Because I sampled an even 65 records per category, the corpus does not reflect how often each issue really happens, and the construction category in particular is thin and blurs into permits. Almost every record routes to Permits, Licenses and Inspections, so there is not much to learn about routing. Finally, there were no near-duplicate records to reconcile, since that check came back at 0%.